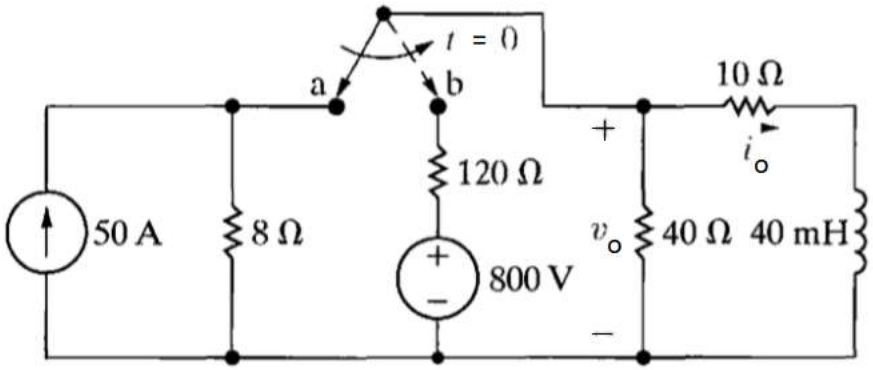
Name & Surname: 11.09.2020

Student Number:

EE231 FINAL

7.37

1) (25pts)



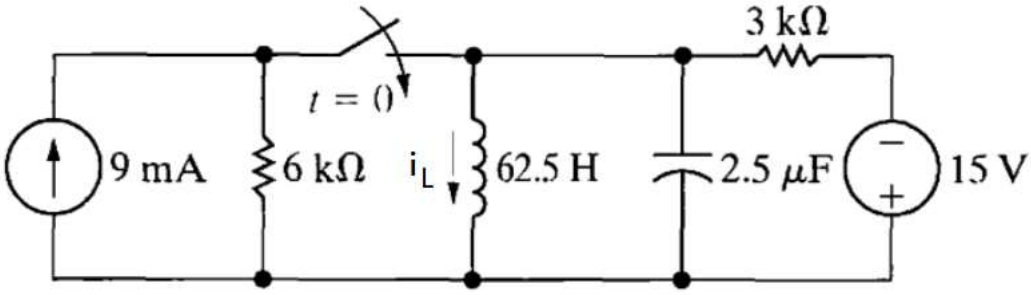
The switch in the circuit shown in the figure has been in position **a** for a long time. At t=0, the switch moves instantaneously to position b.

a) Find the numerical expression for $i_0(t)$ when $t \geq 0$.

b) Find the numerical expression for $v_0(t)$ for $t \geq 0^+$.

8.32

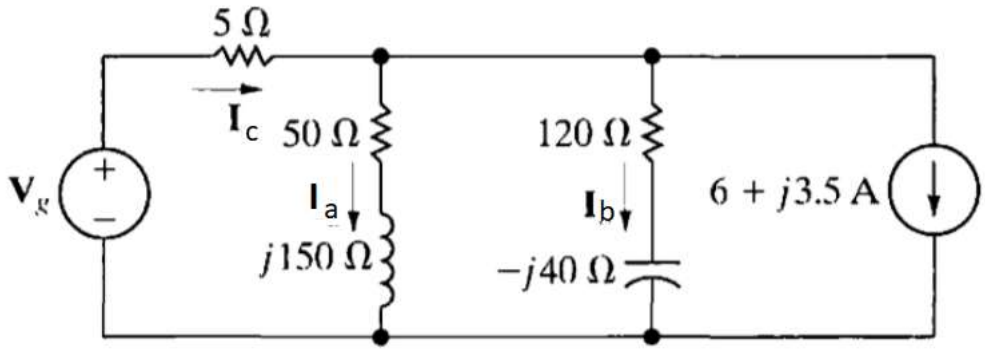
2) (25pts)



The switch in the circuit has been open a long time before closing at t=0. Find $i_L(t)$ for $t \geq 0$.

3) (25pts)

9.33



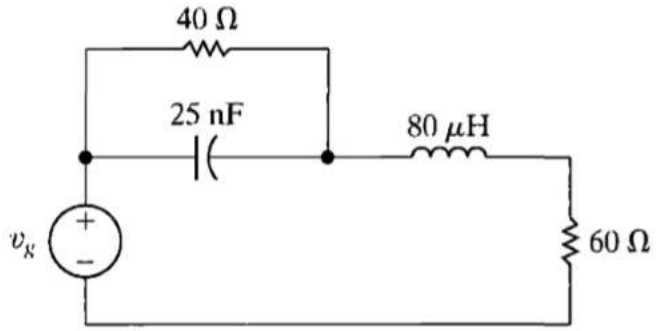
The phasor current $\mathbf{I_a}$ in the circuit shown is $2\angle 0°$ A.

a) Find $\mathbf{I_b}$, $\mathbf{I_c}$, and $\mathbf{V_g}$.

b) If ω = 800 rad/s, write the expressions for $i_b(t)$, $i_c(t)$ and $v_g(t)$.

4) (25pts)

10.18



a) (10pts) Find the average power, the reactive power, and the apparent power supplied by the voltage source in the circuit if $v_g = 40 \cos 10^6 t$ V.

b) (10pts) Check your answer in (a) by showing $P_{dev} = \sum P_{abs}$.

c) (5pts) Check your answer in (a) by showing $Q_{dev} = \sum Q_{abs.}$

Good Luck!